

Coursre: Automata

Lecturer: Pun Solita

Assignment01

Team Member Group 03

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Automata - Practice 1

1. Sets

- a. $A = \{\text{odd numbers between 8 and 14}\}$

List the elements of set A

$$A = \{9, 11, 13\}$$

- b. $B = \{\text{square numbers between 10 and 50}\}$

List the elements of set B

$$B = \{16, 25, 36, 49\}$$

- c. $C = \{\text{multiples of 3 between 1 and 31}\}$

$D = \{\text{multiples of 4 between 1 and 31}\}$

List the members of both set C and set D

$$C = \{3, 6, 9, 12, 15, 18, 21, 24, 27, 30\}$$

$$D = \{4, 8, 12, 16, 20, 24, 28\}$$

- d. Circle the best description of $C \cup D$

1. C but not D

2. D but not C

3. C and D

☒ 4. C or D or both

- e. Circle the best description of $C \cap D$

1. C but not D

2. D but not C

☒ 3. C and D

4. C or D or both

- f. Circle the best description of C'

2. C

2. D

③ Not C

4. Not D

g. $A = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$

$B = \{7, 8, 9\}$

List the members of $A \cap B$

$A \cap B = \{7, 8, 9\}$

h. $A = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$

$B = \{7, 8, 9\}$

List the members of $A \cup B$

$A \cup B = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$

i. $E = \{11, 12, 13, 14, 15, 16, 17\}$

$F = \{12, 14, 16, 18\}$

Find the set $E \cap F$

$E \cap F = \{12, 14, 16\}$

j. $U = \{20, 21, 22, 23, 24, 25, 26, 27\}$

$A = \{20, 23, 24, 25, 26\}$

Find the A'

$A' = \{21, 22, 27\}$

k. $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14\}$

$A = \{1, 7, 10, 13\}$

$B = \{3, 5, 7, 9, 11, 13\}$

1. List the members of $A' \cap B'$

$A' \cap B' = \{2, 4, 8, 12, 14\}$

2. List the members of $A \cup B'$

$$A \cup B' = \{1, 2, 4, 7, 8, 10, 12, 13, 14\}$$

1. $U = \{13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23\}$

$A = \{13, 17, 19, 21, 23\}$

$B = \{17, 18, 19, 20, 21, 22, 23\}$

1. List the members of $A' \cup B'$

$$A' \cup B' = \{13, 14, 15, 16, 18, 20, 22\}$$

2. List the members of $A' \cap B'$

$$A' \cap B' = \{14, 15, 16\}$$

3. List the members of $A' \cap B$

$$A' \cap B = \{18, 20, 22\}$$

4. List the members of $A \cup B'$

$$A \cup B' = \{13, 14, 15, 16, 17, 19, 21, 23\}$$

2. Sequence

a. Let $A = \{1, 2\}$ and $B = \{x, y, z\}$. Write out all the ordered pairs that comprise the Cartesian product $A \times B$

$$A \times B = \{(1, x), (1, y), (1, z), (2, x), (2, y), (2, z)\}$$

b. Given the set $A = \{1, 2, 3, 4\}$, list all elements of the powerset $P(A)$

$$P(A) = 2^n = 2^4 = 16$$

$$P(A) = \{\emptyset, 1, 2, 3, 4, (1, 2), (1, 3), (1, 4), (2, 3), (2, 4), (3, 4), (1, 2, 3), (1, 2, 4), (1, 3, 4), (2, 3, 4), (1, 2, 3, 4)\}$$

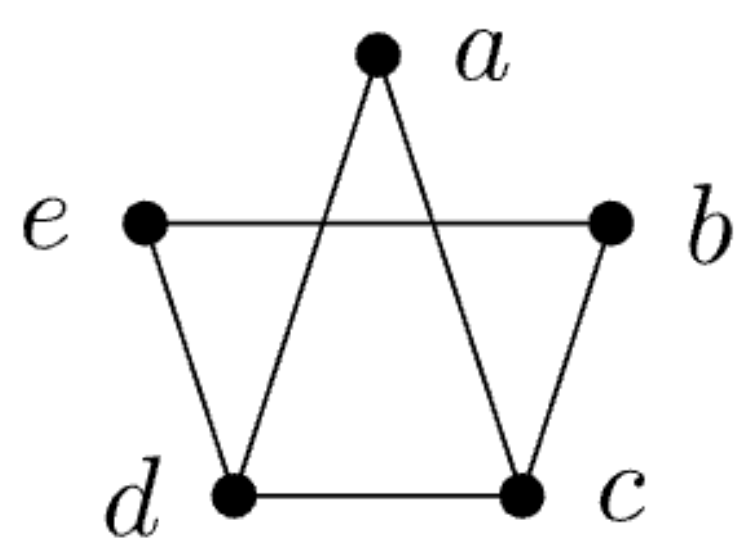
3. Graph

a. If 10 people each shake hands with each other, how many handshakes took place? Draw the graph:

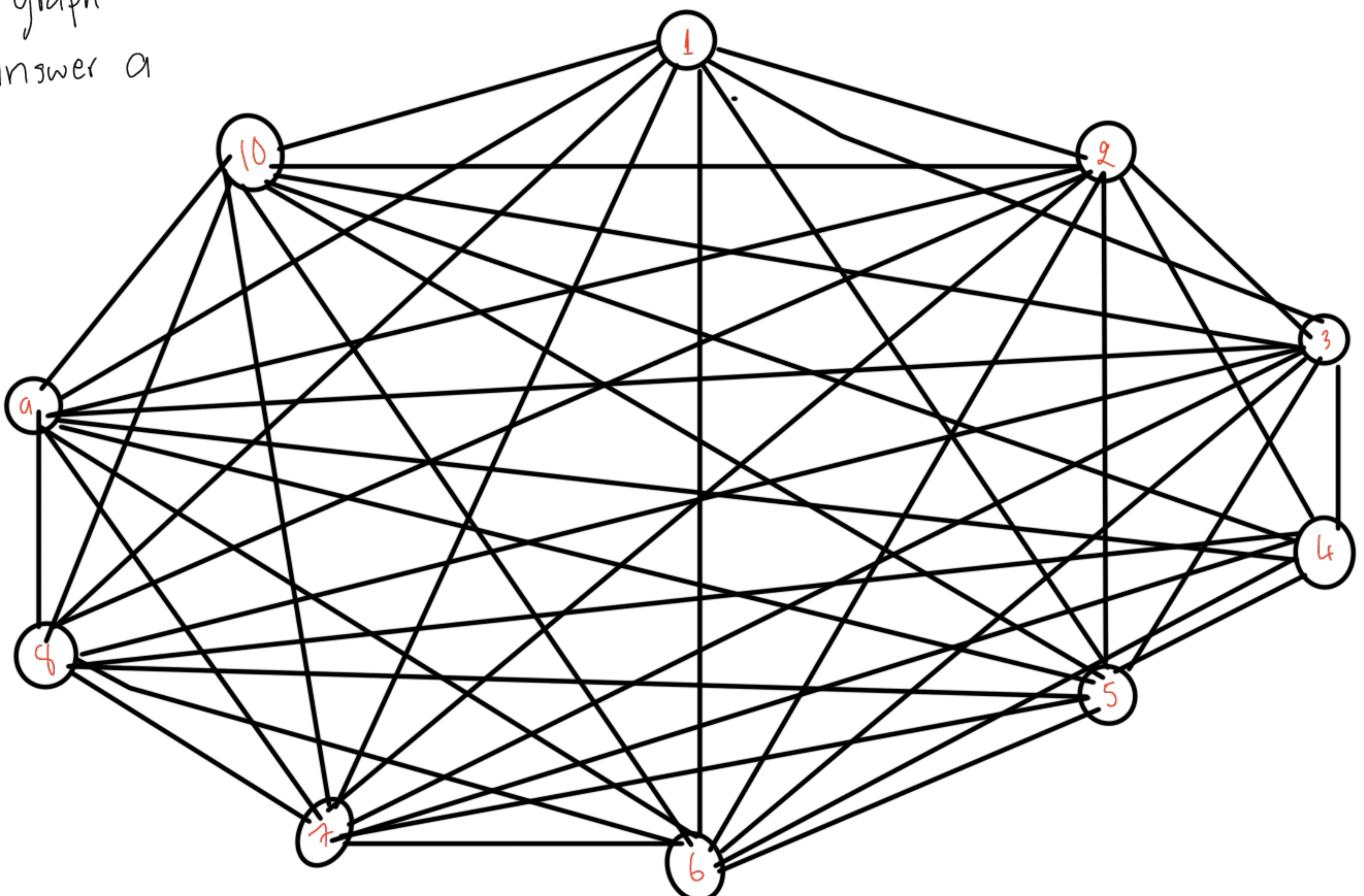
b. Are the two graphs below equal? If not, explain.

Graph 1: $V = \{a, b, c, d, e\}$, $E = \{\{a, b\}, \{a, c\}, \{a, e\}, \{b, d\}, \{b, e\}, \{c, d\}\}$.

Graph 2:



3, graph
answer a



Answer B

These 2 graph doesn't equal to each other because
2 graph doesn't have the edges that connect vertices $\{a, b\}$ and $\{a, e\}$ ---
like graph 1 does.